HAWLER MEDICAL UNIVERSITY | COLLEGE OF MEDICINE

Water and Electrolyte Balance by the Kidneys LG1 GUE AY 26-27

Assist. Prof. Dr. Sherwan R Shal MBChB MSc PhD (Medicine , Medical Physiology & CT)

Core mechanisms • labeled diagrams • applied examples • SBA review

Learning outcomes & lecture map

- Define water balance, osmolality, reabsorption and secretion.
- Explain how nephron segments handle water and major ions.
- Describe ADH, the medullary gradient, RAAS and aldosterone.
- Predict renal responses to dehydration, high water intake and potassium load.

Foundations · 10 min

Mechanisms · 35 min

Application/SBA · 15 min

Guyton & Hall, 15th ed.; Boron & Boulpaep, 4th ed.

Water Balance

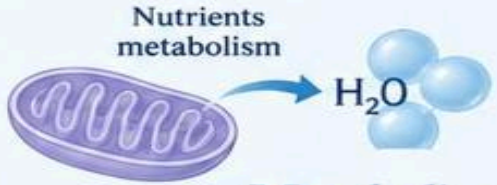
Water Intake (≈ 2,500 mL/day)



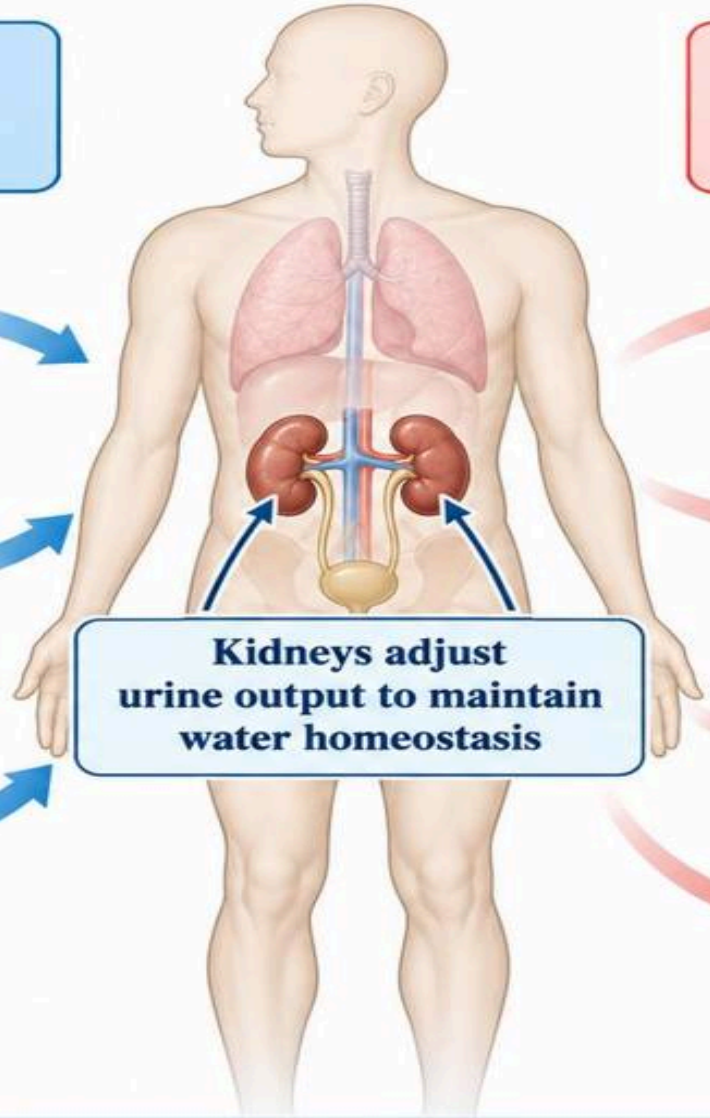
Drinks
≈ 1,200 mL/day



Food
≈ 1,000 mL/day



Metabolic water
≈ 300 mL/day



Kidneys adjust urine output to maintain water homeostasis

Water Output (≈ 2,500 mL/day)



Urine
≈ 1,500 mL/day



Skin (sweat)
≈ 500 mL/day



Lungs
≈ 400 mL/day



Feces
≈ 100 mL/day

Water intake ≈ water output in steady state.

The principle: intake must equal output

Water balance

Daily intake = drinking + food + metabolic water.
Daily output = urine + skin + lungs + stool.

Electrolyte balance

Urinary excretion adjusts to dietary intake.
The kidneys conserve what is needed and eliminate excess.

RENAL EXCRETION = FILTRATION – REABSORPTION + SECRETION

Example: during dehydration, a healthy kidney reduces urine volume and increases urine concentration.

Essential definitions: compartments & tonicity

Total body water (TBW)

All water in the body; ≈60% of body mass in a typical young adult male.

Intracellular fluid (ICF)

Fluid inside cells; approximately ⅔ of TBW.

Extracellular fluid (ECF)

Fluid outside cells; approximately ⅓ of TBW.

Osmolality

Number of dissolved particles per kg of water (mOsm/kg).

Tonicity

Effective osmotic force that changes cell volume.

Major ECF cation: Na^+ | Major ICF cation: K^+

Nephron segments: where reabsorption occurs

01 Glomerulus

Filtration

02 Proximal tubule

≈65% Na^+ and water

03 Descending limb

Water exits

04 Thick ascending limb

Na^+–K^+–$2Cl^-$; no water

05 Distal tubule

Fine-tunes NaCl / Ca^{2+}

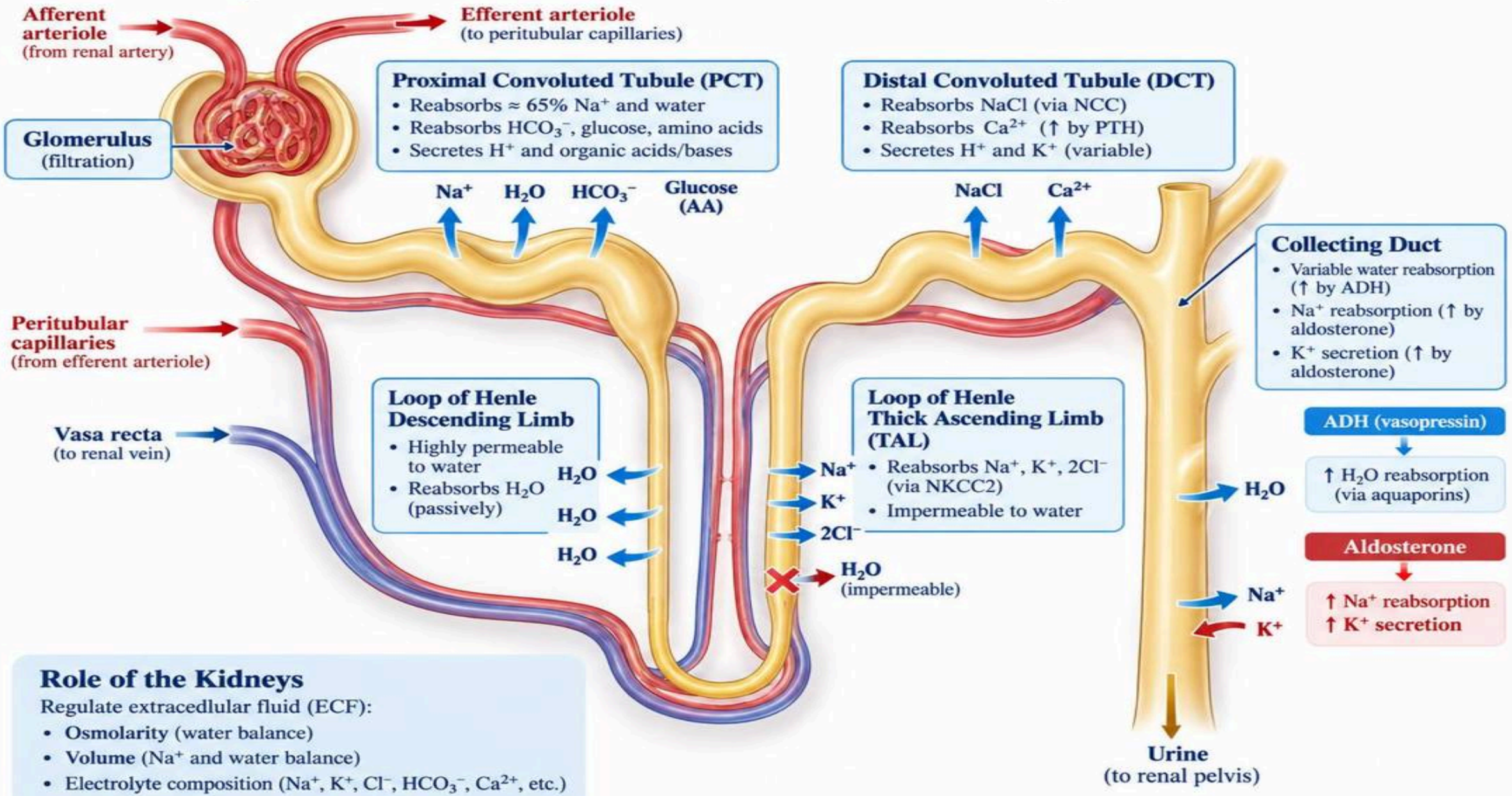
06 Collecting duct

ADH and aldosterone

Most filtered fluid is reabsorbed early; hormones control the final adjustment in the distal nephron.

Guyton & Hall, 15th ed.; Boron & Boulpaep, 4th ed.

Nephron in Water & Electrolyte Balance



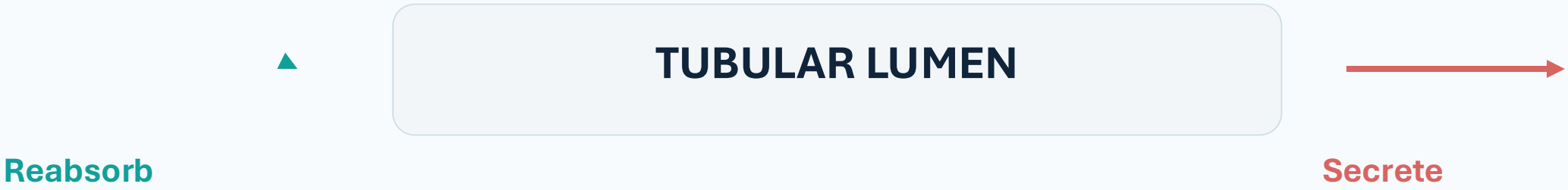
How the nephron transports water and solutes

Reabsorption

Tubule → interstitium → peritubular blood
Example: Na^+ and water return to circulation.

Secretion

Peritubular blood → tubule
Example: K^+ is added to final urine.

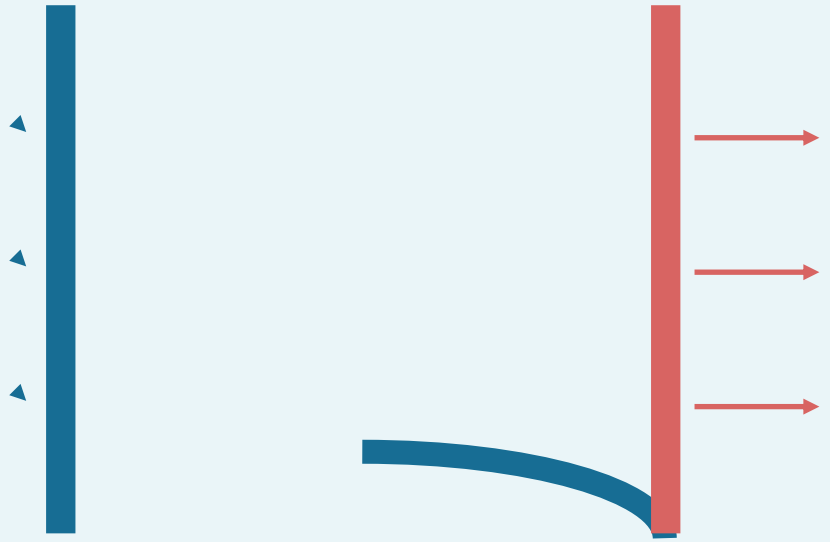


Water moves by osmosis; active Na^+ transport creates many of the gradients that drive reabsorption.

Guyton & Hall, 15th ed.; Boron & Boulpaep, 4th ed.

Countercurrent multiplier: build the medullary gradient

CORTEX ≈300 mOsm/kg



INNER MEDULLA up to ≈1200 mOsm/kg

The vasa recta preserve the gradient; urea recycling contributes especially to the inner medulla.

Guyton & Hall, 15th ed.; Boron & Boulpaep, 4th ed.

ADH: the final control of water excretion

↑ Plasma osmolality or ↓ effective circulating volume

↓

Hypothalamic sensing → posterior pituitary releases ADH

↓

ADH binds V_2 receptor on collecting-duct principal cell

↓

Aquaporin-2 channels inserted in luminal membrane

↓

↑ Water reabsorption → ↓ urine volume, ↑ urine osmolality

ADH primarily regulates water; it does not directly “pump sodium” across the collecting duct.

Guyton & Hall, 15th ed.; Boron & Boulpaep, 4th ed.

ADH Action

1 **Increased plasma osmolality or low blood volume stimulates ADH release**

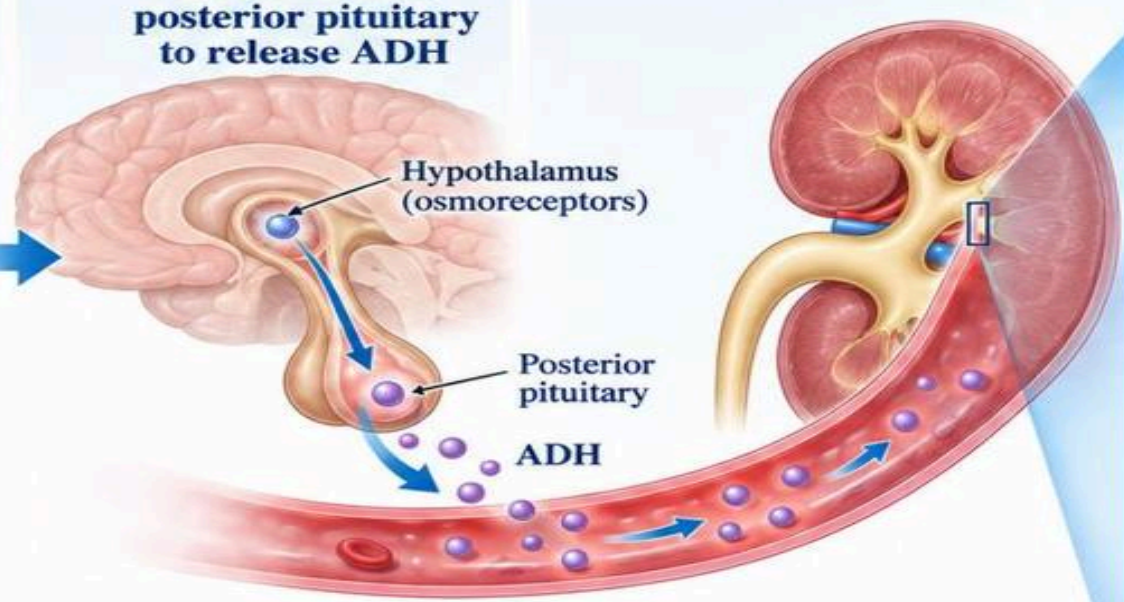
↑ **Plasma osmolality** (e.g., dehydration)

and/or

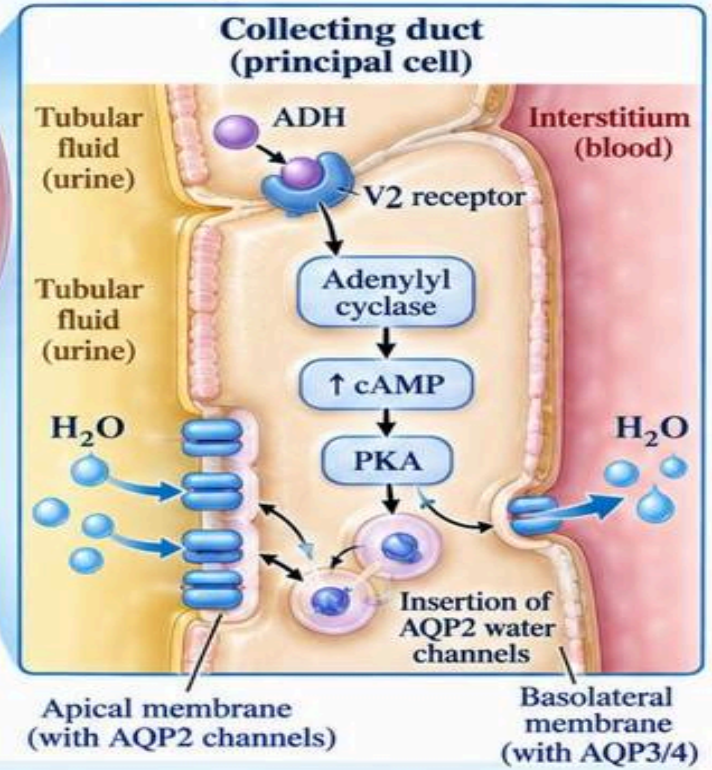
↓ **Blood volume** (e.g., hemorrhage)

2 **Hypothalamus senses the change and stimulates posterior pituitary to release ADH**

3 **ADH travels in the blood to the kidneys**



4 **ADH binds to V2 receptors on principal cells in the collecting duct**



5 **Increased water reabsorption leads to concentrated urine and reduced urine volume**

↑ **Water permeability of collecting duct** (insertion of AQP2)

↑ **Water reabsorption** (from tubular fluid into interstitium/blood)



Collecting duct
(tubular fluid → interstitium)



Concentrated urine (high osmolality)

Reduced urine volume

Dilute vs concentrated urine

After a large water intake

↓ ADH → fewer AQP2 channels
More water stays in the tubule
Large volume of dilute urine

During water deprivation

↑ ADH → more AQP2 channels
Water exits into hyperosmotic medulla
Small volume of concentrated urine

Urine osmolality is dynamic; concentration requires both ADH and an intact medullary gradient.

Example: a person who drinks several litres of water produces much more dilute urine than after overnight fluid restriction.

Sodium balance: ECF volume and arterial pressure

- Na^+ is the predominant extracellular cation.
- The kidneys regulate total-body Na^+ content by adjusting excretion.
- The proximal tubule reabsorbs most filtered Na^+; distal segments fine-tune it.
- Retaining Na^+ tends to expand ECF volume, provided water is available.

Key distinction: total-body Na^+ mainly affects ECF volume; plasma [Na^+] mainly reflects water balance relative to solute.

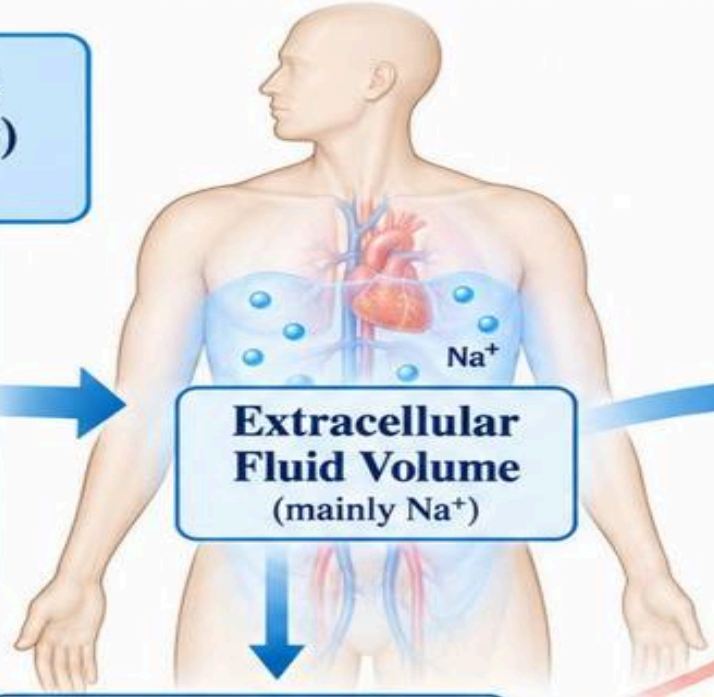
Guyton & Hall, 15th ed.; Boron & Boulpaep, 4th ed.

Sodium Balance

Sodium Intake
(≈ 100–200 mEq/day)
(≈ 2–5 g Na^+/day)



Dietary sodium enters the body

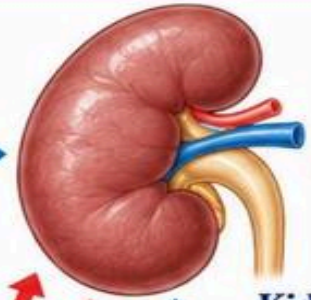


Extracellular Fluid Volume
(mainly Na^+)

Determines blood volume and helps regulate blood pressure



Renal Sodium Excretion
(≈ 100–200 mEq/day)
(≈ 2–5 g Na^+/day)



Kidneys adjust sodium excretion to match intake over time

Urinary Sodium Output
(≈ 100–200 mEq/day)
(≈ 2–5 g Na^+/day)

Regulation of Renal Sodium Excretion

RAAS (Angiotensin II / Aldosterone)
↑ Na^+ reabsorption
(↑ retention)



Activated by low ECF volume / low blood pressure

Sympathetic Nervous System
↑ Na^+ reabsorption
(↑ retention)



Activated by low blood pressure or stress

Natriuretic Peptides (ANP, BNP)
↑ Na^+ excretion
(↓ reabsorption)



Released with increased atrial/ventricular stretch (high ECF volume)

Pressure Natriuresis
↑ Na^+ excretion



Higher arterial pressure increases renal Na^+ excretion (independent of hormones)

In steady state: Sodium intake = Sodium output
(ECF volume and blood pressure remain stable)

RAAS and aldosterone: conserve sodium

↓ Renal perfusion / ↓ NaCl delivery

Triggers renin release

Renin → angiotensin I → angiotensin II

Ang II constricts vessels and stimulates aldosterone

Aldosterone at distal nephron

↑ ENaC and Na^+/K^+-ATPase activity

Net response

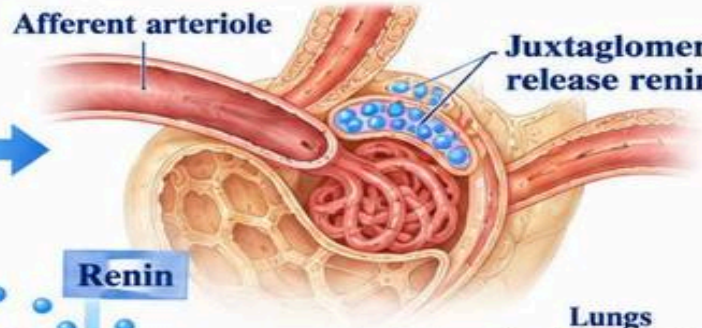
↑ Na^+ reabsorption; ↑ K^+ secretion

ANP, released with atrial stretch, generally promotes natriuresis (urinary Na^+ loss).

RAAS Regulation

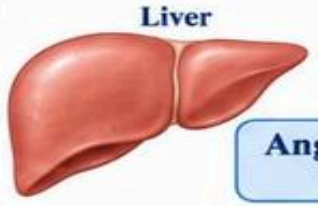
Renin-Angiotensin-Aldosterone System

Triggers (↓ renal perfusion)
- Low blood pressure
- Low renal perfusion
- Low NaCl at macula densa
- Sympathetic stimulation (β_1-adrenergic)



Stimulated by:
- ↓ Pressure in afferent arteriole
- ↓ NaCl delivery to macula densa
- ↑ Sympathetic activity (β_1)

Renin



Angiotensinogen (in plasma) → **Angiotensin I** (in blood) → **ACE** (Lungs, endothelial cells) → **Angiotensin II** (in blood)

Vasoconstriction



- ↑ Systemic vascular resistance (↑ BP)
- Constricts efferent arteriole (maintains GFR)

Aldosterone release



- Angiotensin II stimulates aldosterone secretion

ADH release and thirst



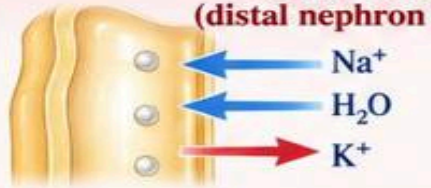
- Stimulates ADH release (from posterior pituitary)
- Stimulates thirst (in hypothalamus)

↑ Proximal Na^+ reabsorption



- Directly increases Na^+ reabsorption in proximal tubule
- Increases Na^+/H^+ exchanger activity

Actions of aldosterone
(distal nephron / collecting duct)



- ↑ Na^+ reabsorption (ENaC, Na^+/K^+ ATPase)
- ↑ Water retention (follows Na^+)
- ↑ K^+ secretion

↑ Blood pressure and ↑ ECF volume
(restoration of renal perfusion)

Potassium balance: secretion is the key adjustment

Why K^+ matters

The major intracellular cation; essential for resting membrane potential, nerve and muscle function.

Where is K^+ controlled?

Most filtered K^+ is reabsorbed earlier; variable secretion occurs in the late distal tubule and collecting duct.

High plasma K^+ → **↑ Aldosterone** → **↑ K^+ secretion**

Example: after a potassium-rich meal, intact kidneys increase urinary K^+ excretion.

Guyton & Hall, 15th ed.; Boron & Boulpaep, 4th ed.

Major electrolyte handling: summary

Ion	Main renal regulation	Physiological importance
Na^+	Reabsorption; RAAS/ANP	ECF volume and pressure
K^+	Distal secretion; aldosterone	Membrane excitability
Ca^{2+}	Reabsorption; PTH	Bone, contraction, signaling
HCO_3^- / H^+	Reabsorb HCO_3^-; secrete H^+	Acid–base balance

Phosphate and Mg^{2+} are also regulated by the kidneys; this lecture emphasizes Na^+, K^+, Ca^{2+} and acid–base ions.

Guyton & Hall, 15th ed.; Boron & Boulpaep, 4th ed.

Apply the concepts: 3 physiological scenarios

A. Heavy sweating, little water intake

↑ osmolality ± ↓ effective volume → ↑ ADH → water conservation

B. Large water intake in a healthy person

↓ ADH → high-volume dilute urine

C. Increased dietary K^+

↑ distal K^+ secretion, supported by aldosterone

Ask students: What changes first: the stimulus, hormone, transport pathway, or urine response?

SBA review + core references

1. The nephron segment impermeable to water is:

A. Proximal tubule B. Descending limb C. Thick ascending limb D. Collecting duct

Answer: C

2. ADH increases water permeability mainly through:

A. ENaC B. AQP2 C. NKCC2 D. Na^+/K^+-ATPase

Answer: B

3. Aldosterone usually increases renal excretion of:

A. Na^+ B. K^+ C. Water only D. Glucose

Answer: B

Further reading: Hall & Hall, Guyton and Hall Textbook of Medical Physiology, 15th ed. (2026), renal chapters; Boron & Boulpaep Medical Physiology, 4th ed. (2022); NCBI Bookshelf renal physiology; Merck Manual (2026).